LUCIAN URITU

Maastricht, Netherlands | lucian@uritu.net | +31 6 87611935 | GitHub Lucian Uritu

Computer Science student developing desktop software for Maastricht University Press, with hands-on experience in Java networking, API integration, and parallel computing.

PROFESSIONAL EXPERIENCE

Maastricht University, Maastricht, Netherlands — Apr 2026 - Present
Student Assistant - Software Development

- Proposed and took ownership of the end-to-end design and development of an authoring platform for Maastricht University Press.
- Developed a desktop application using JavaScript, Node.js, HTML/CSS and the GitHub API so non-technical authors can create, preview, manage and publish Markdown educational content.
- Translate feedback from researchers and faculty into features; prepare documentation and user guides for a planned first production rollout to 12 users in November 2026.
- Used AI coding assistants for code generation, debugging, refactoring, and documentation, while reviewing generated code.

SELECTED PROJECTS

Unblocking the City — Jan 2026 - Jun 2026
GitHub Code

- Implemented MPI inter-process communication for distributed urban traffic simulation with SUMO on HPC clusters as part of a six-person team.
- Evaluated graph partitioning, scalability and bottlenecks; parallel Monte Carlo experiments achieved up to 14.9x speedup on 32 CPU cores.

Gemini-Lite Protocol — Oct 2025 - Dec 2025
GitHub Code

- Developed a modular Gemini protocol server and client in Java with proxying, file handling and custom protocol extensions.
- Implemented networking and request/response parsing, and integrated comprehensive testing.

Helsinki Route Planner — Jan 2025 - Jun 2025
GitHub Code

- Built a JavaFX public transport application for Helsinki with real-time journey planning using Dijkstra's shortest-path algorithm and transit heatmap visualization.
- Simulated 100,000 journeys across 662 Helsinki transit stops to evaluate route accessibility and identify underutilized infrastructure.
- Analyzed 8 closure candidates and identified 3 stops whose removal resulted in an average additional travel delay of ≤1 minute in the simulation.

EDUCATION

Maastricht University, Maastricht, The Netherlands — Sep 2024 - Jul 2027 (expected)
Bachelor of Science in Computer Science

SKILLS & OTHER

Programming Languages Java, Python, JavaScript, C, HTML/CSS

Frameworks and tools Node.js, Electron, JavaFX, Git, GitHub API, Linux/Unix

Parallel computing MPI, CUDA, OpenMP, SUMO

Skills Distributed Systems Programming, OOP, Databases, Web Development, Network Programming, Embedded Programming, AI-assisted Development

Languages Romanian (native), English (C2), Russian (conversational)